

SEMESTER END EXAMINATIONS – JANUARY 2026

Program	: B.E. – Information Science and Engineering	Semester	: V
Course Name	: Artificial Intelligence	Max. Marks	: 100
Course Code	: 23IS52	Duration	: 3 Hrs

Instructions to the Candidates:

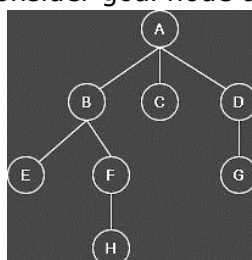
- Answer one full question from each unit.

UNIT - I

- Explain the four major dimensions used to define Artificial Intelligence, namely:
(i) Acting Humanly, (ii) Thinking Humanly, (iii) Thinking Rationally, and (iv) Acting Rationally. CO1 (08)
 - Describe the structure of a simple intelligent agent with a neat diagram. CO1 (06)
 - Drones are used in smart agriculture for crop monitoring, pesticide spraying, and detecting plant health issues. Formulate the PEAS description for the drone. CO1 (06)
- Illustrate with a schematic diagram the working principle of utility-based agent with a pseudocode. CO1 (08)
 - Describe the functioning of a Learning Agent. Illustrate with a diagram. CO1 (06)
 - An online chess-playing system competes against human players on the internet. The system receives full-board information before each move. Describe the environment in terms of observability, agents, and whether it is deterministic or stochastic. CO1 (06)

UNIT - II

- Compare Uninformed and Informed Search Strategies in solving relational world problems. CO2 (05)
 - Formulate the Vacuum Cleaner World as a problem for a problem-solving agent. Define its states, initial state, goal state, action cost, transition model, and performance measure. Write a suitable diagram of the environment. CO2 (08)
 - With an example graph, Perform Uniform Cost Search (UCS) and determine the optimal path to the goal. Analyze why UCS is complete and optimal. CO2 (07)
- Given the following tree: consider goal node as G CO2 (05)



- i) Perform BFS starting from node A. Write the order of visited nodes.
- ii) Perform DFS starting from node A. Write the order of traversal.
- iii) State the time complexity and efficiency of BFS and DFS.
- b) Apply the A* search algorithm on the following graph where each edge has a cost and each node has a heuristic $h(n)$. Compute $f(n) = g(n) + h(n)$ at every step. CO2 (08)

1	2	3
4	5	6
	7	8

Start state

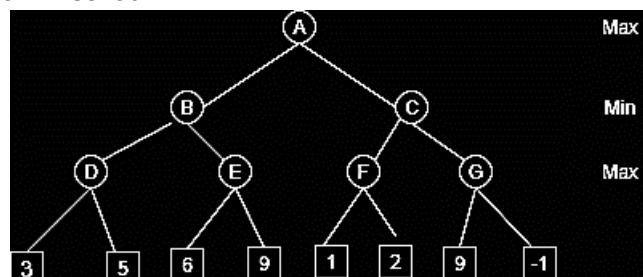
1	2	3
4	5	
7	8	6

Goal State

- c) With an example, analyse how bidirectional search is used for solving real world problem with example. CO2 (07)

UNIT - III

5. a) Write and interpret the propositional logic to represent the knowledge derived by each of the following sentences for Wumpus world problem: CO3 (08)
 - i) Square (1, 2) is breezy.
 - ii) No stench in square (2, 1).
 - iii) Glitter in (2,4) but not in (2,3)
 - iv) Stench in (1,3) and no stench in (1,2).
- b) Explain the PEAS and characterization of Wumpus world. CO3 (06)
- c) Explain the minimax algorithm. Solve the following game tree problem using minimax method. CO3 (06)



6. a) Explain the syntax and semantics of propositional logic with examples. CO3 (08)
- b) A Knowledge Base contains the following information. Describe the possible inferences drawn from the given knowledge base. CO3 (06)
 - Square (1,1) has no breeze.
 - Square (1,1) has no stench.
 - Square (1,2) is breezy.
 - Square (2,1) is breezy.
 - Square (1,2) has no stench.
 - There is exactly one Wumpus.
- c) Differences between minimax and alpha beta pruning methods. CO3 (06)

UNIT- IV

7. a) Differentiate between Propositional Logic and First-Order Logic. Explain why FOL is more expressive. CO4 (10)
- b) Describe the Forward Chaining and Backward Chaining algorithms used in FOL inference. CO4 (10)

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| 8. | a) | Explain the process of Unification in FOL with examples. Why is it essential for inference? | CO4 | (10) |
| | b) | Discuss the process of Knowledge Engineering in FOL. How can real-world knowledge be modelled using FOL statements? | CO4 | (10) |

UNIT - V

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| 9. | a) | Can machines really think? Justify the statement with examples. | CO5 | (10) |
| | b) | Discuss the PDDL description of The simple spare tire problem. | CO5 | (10) |
| 10. | a) | Discuss the PDDL description of an air cargo transportation planning problem. | CO5 | (08) |
| | b) | Discuss some the fresh problems posed by AI. | CO5 | (06) |
| | c) | Can machines really think? Discuss. | CO5 | (06) |
